

Supplementary Material for TokBench

Junfeng Wu*
wjf5203@gmail.com
Huazhong University of Science and
Technology
Wuhan, China

Zhihao Xie
zhxie17@hust.edu.cn
Huazhong University of Science and
Technology
Wuhan, China

Xudong Xie
xdxie@hust.edu.cn
Huazhong University of Science and
Technology
Wuhan, China

Dongliang Luo*
ldl@hust.edu.cn
Huazhong University of Science and
Technology
Wuhan, China

Yuanhao Wang
yhwang7@hust.edu.cn
Huazhong University of Science and
Technology
Wuhan, China

Yuliang Liu
ylliu@hust.edu.cn
Huazhong University of Science and
Technology
Wuhan, China

Weizhi Zhao
zhaoweizhi@hust.edu.cn
Huazhong University of Science and
Technology
Wuhan, China

Junyi Li
lji1308598378@gmail.com
Huazhong University of Science and
Technology
Wuhan, China

Xiang Bai
xbai@hust.edu.cn
Huazhong University of Science and
Technology
Wuhan, China

Overview

This supplementary material provides additional details and experimental results to complement the main paper. Section A covers dataset licenses, difficulty rating definitions, and tokenizer model specifications. Section B presents multi-resolution evaluation results for text and face reconstruction, video evaluation at 480 resolution, analysis of the relationship between reconstruction and generation performance, and additional metric validation experiments. Section C describes ablation studies on the impact of training data composition. Section D provides qualitative reconstruction comparisons at 256 and 1024 resolutions.

A Dataset Details

A.1 Dataset Licenses

TokBench is curated from publicly available datasets for research purposes. We list the license information of all source datasets in Table 1. All datasets are used in compliance with their respective licenses and are intended solely for non-commercial academic research. We do not redistribute the original images or videos; our benchmark provides only annotations and evaluation scripts, with instructions to download source data from their official repositories.

A.2 Difficulty Rating

We classify target instances into three difficulty levels (Small, Medium, Large) based on their relative scale, as detailed in Table 2. The lower bound for each resolution is determined by the minimum pixel requirement for recognizable content.

For text, we assume at least 5×5 pixels are needed to represent a single character. At resolution L , instances with character-level scale below $5/L$ are excluded from evaluation. For example, at 256 resolution, the minimum character scale is $5/256 \approx 0.02$. For faces, we empirically set 25 pixels as the minimum for recognizable identity based on visualization and analysis of the resize upper

Table 1: License information for source datasets used in TokBench.

Dataset	Type	License
<i>Image – Text</i>		
Total-Text	Scene text	BSD-3-Clause
TextOCR	Scene text	CC BY 4.0
ICDAR 2013	Scene text	Research only
ICDAR 2015	Scene text	Research only
SROIE	Receipt	MIT
CORD	Receipt	CC BY 4.0
DocVQA	Document	MIT
InfographicVQA	Document	MIT
<i>Image – Face</i>		
WFLW	Face	Research only [†]
<i>Video – Text</i>		
ICDAR 2015 Video	Video text	Research only
DSTextV2	Video text	Research only
<i>Video – Face</i>		
VideoMME	Video	Research only
MVBench	Video	CC BY 4.0 [‡]
MMBench-Video	Video	CC BY-NC-SA 4.0

[†] Images sourced from WIDER Face (non-commercial research only).

[‡] Contains videos from multiple sources with varying licenses.

bound, yielding a minimum face scale of $25/256 \approx 0.10$ at 256 resolution.

For video evaluation, since most videos follow a 16:9 aspect ratio and we resize the shorter edge to the target resolution (resulting in longer edges around 500 pixels), we adopt slightly more lenient scale thresholds compared to image evaluation to accommodate these aspect ratio differences.

A.3 Tokenizer Selection

We detail the specific model versions used in our evaluation. For continuous VAEs, we employ VA-VAE [25], SDXL [16], SD3.5 [4],

Table 2: Difficulty rating scale ranges. The lower bound for each resolution is determined by the minimum pixel requirement for recognizable faces. Instances below the Small threshold are excluded from evaluation at that resolution.

Type	Cat.	Res.	Small	Medium	Large
Image	Text	256	0.02 ~ 0.03	0.03 ~ 0.04	0.04 ~ 1.00
		512	0.01 ~ 0.02	0.02 ~ 0.03	0.03 ~ 1.00
		1024	0.005 ~ 0.01	0.01 ~ 0.02	0.02 ~ 1.00
	Face	256	0.10 ~ 0.20	0.20 ~ 0.30	0.30 ~ 1.00
		512	0.05 ~ 0.10	0.10 ~ 0.20	0.20 ~ 1.00
		1024	0.02 ~ 0.05	0.05 ~ 0.10	0.10 ~ 1.00
Video	Text	256	0.01 ~ 0.02	0.02 ~ 0.03	0.03 ~ 1.00
		480	0.005 ~ 0.01	0.01 ~ 0.02	0.02 ~ 1.00
	Face	256	0.05 ~ 0.10	0.10 ~ 0.20	0.20 ~ 1.00
		480	0.02 ~ 0.05	0.05 ~ 0.10	0.10 ~ 1.00

and FLUX [8] VAEs obtained from their official HuggingFace repositories, along with the 32× downsampling DC-AE with 32 latent dimensions from SANA [23].

For discrete tokenizers, we adopt the ImageNet-trained VQ-GAN [5] (F16, codebook size 16,384) as the baseline. LlamaGen [18] is evaluated in both F16 and F8 variants. VAR [19] uses the largest VAR-d36 model (2.3B parameters). TiTok [26] uses the TiTok-L-32 configuration, representing each image with 32 tokens. For OpenMAGVIT2 [11], we evaluate both F16 and F8 models trained on ImageNet, as well as an F16 model pretrained on 100M images with a codebook size of 262,144. MaskBit [22] uses the 12-bit variant. OmniTokenizer [21] uses the recommended imagenet_k600 version. FlexTok [1] uses the version trained on the DFN dataset [6]. UniTok [12] uses the released checkpoint with 4 sub-codebooks.

B Detailed Evaluation Results

B.1 Multi-Resolution Text Evaluation

Table 3 presents text reconstruction results across three resolutions (256, 512, 1024). Most tokenizers show progressive improvement with increasing resolution, as higher resolutions preserve more pixel-level detail for text characters. However, small-scale text remains challenging across all resolutions: even at 1024, discrete tokenizers with F16 downsampling achieve less than 5% T-ACC on small-scale text. The performance gap between continuous and discrete tokenizers widens significantly at higher resolutions. At 1024 resolution, FLUX’s VAE achieves T-NED comparable to the resize upper bound, while most discrete tokenizers still lag considerably. Notably, since many original text images exceed 1024 pixels, the resize baseline itself cannot achieve perfect T-ACC at this resolution. Furthermore, discrepancies between conventional metrics and T-ACC become more pronounced at higher resolutions: for instance, at 512 resolution LlamaGen and TokenFlow show similar LPIPS scores but differ substantially in T-ACC, and at 1024 resolution UniTok and Open-MAGVIT2 exhibit similar rFID yet diverge significantly in text reconstruction quality.

B.2 Multi-Resolution Face Evaluation

Table 4 presents face reconstruction results across three resolutions. Since most face images do not exceed 1024 pixels, the resize baseline at 1024 resolution preserves nearly identical facial details, yielding F-Sim close to 1.0. At this resolution, continuous VAEs from SD3.5 and FLUX achieve near-perfect performance, while even the best discrete tokenizers only reach approximately 0.5 F-Sim for small-scale faces. This highlights a persistent gap between discrete and continuous methods for fine-grained facial identity preservation at small scales. At lower resolutions, performance degrades substantially for all methods, with small-scale faces proving particularly challenging: at 256 resolution, most F16 discrete tokenizers achieve F-Sim below 0.15 on small-scale faces. Compared to text reconstruction, face reconstruction generally approaches the resize upper bound more closely, likely because existing tokenizers have been exposed to extensive facial data during training.

B.3 Video Evaluation at 480 Resolution

Table 5 presents video reconstruction results at 480 resolution, complementing the 256-resolution results in the main paper. At this higher resolution, all tokenizers show improved performance, with continuous VAEs benefiting more substantially. Hunyuan-Video and CogVideoX VAEs achieve strong text and face reconstruction at this resolution, narrowing the gap with the resize upper bound. The discrete Cosmos-VAE variants, however, continue to underperform significantly, particularly on small-scale text where T-ACC remains below 5% even at 480 resolution. Comparing across resolutions, the relative ranking of tokenizers remains largely consistent, confirming that TokBench provides stable and reliable evaluation across different resolution settings.

B.4 Reconstruction vs. Generation Performance

A natural question is whether better reconstruction quality translates to better generation performance. To investigate this, we collect reported reconstruction and class-conditional generation results on the ImageNet benchmark from original publications. Since different tokenizers are trained with diverse generators, datasets, and schedules, these generation results are not directly comparable across methods but provide an indicative reference. We summarize the results in Table 6.

We observe several key findings from this comparison:

General correlation with notable exceptions. Tokenizers with better TokBench scores generally achieve better ImageNet rFID, suggesting that fine-grained reconstruction quality on text and faces is indicative of overall reconstruction fidelity. However, generation performance (gFID) does not always align with reconstruction ranking. For example, VAR ranks fifth in reconstruction but third in generation, highlighting that generative modeling strategy (e.g., next-scale prediction) can compensate for reconstruction limitations.

Reconstruction quality is necessary but not sufficient. TiTok achieves the lowest reconstruction quality on TokBench yet attains mid-ranked gFID, suggesting that generation focuses on plausible sampling rather than detail preservation. Conversely, O-MAGVIT2(F8) achieves excellent reconstruction but its longer token sequences

Table 3: Performance of discrete and continuous tokenizer on TokBench text-set.

Type	Method	Factor	T-ACC(%) $\uparrow$				T-NED(%) $\uparrow$				rFID $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$
			Small	Medium	Large	Mean	Small	Medium	Large	Mean				
Resolution: 256 $\times$ 256														
Discrete	Resize	1 $\times$	86.05	94.65	98.37	93.02	92.98	97.22	99.38	96.53	5.66	0.07	25.40	0.81
	TiTok	1D	0.05	0.06	0.17	0.09	3.04	4.07	5.58	4.23	18.41	0.50	13.80	0.50
	FlexTok	1D	0.55	2.24	18.06	6.95	7.80	14.26	41.21	21.09	11.01	0.31	17.58	0.61
	VQGAN	16 $\times$	0.05	0.12	3.14	1.10	4.34	5.33	15.00	8.22	15.66	0.33	17.17	0.58
	Chameleon	16 $\times$	0.11	0.31	8.19	2.87	4.67	6.65	24.91	12.08	17.60	0.33	17.66	0.59
	LlamaGen	16 $\times$	0.16	0.44	12.25	4.28	5.41	7.50	31.40	14.77	14.23	0.29	18.04	0.61
	VAR	16 $\times$	1.24	6.72	39.26	15.74	10.89	26.26	65.42	34.19	10.30	0.22	19.74	0.66
	MaskBit	16 $\times$	0.16	0.19	7.26	2.54	4.45	5.72	22.37	10.85	17.05	0.37	17.90	0.60
	TokenFlow	16 $\times$	0.28	1.62	18.29	6.73	6.41	12.34	42.64	20.46	11.04	0.26	18.61	0.62
	O-MAGVIT2	16 $\times$	0.34	1.49	20.73	7.52	6.46	12.41	44.10	20.99	10.18	0.25	18.85	0.63
	O-MAGVIT2(pretrain)	16 $\times$	0.80	3.17	27.76	10.58	9.59	19.15	54.02	27.59	9.83	0.24	19.15	0.64
	UniTok	16 $\times$	13.53	42.87	77.35	44.59	38.73	68.51	90.27	65.84	9.21	0.19	20.58	0.68
	OmniTokenizer	8 $\times$	2.14	8.46	51.28	20.63	13.24	30.50	73.67	39.14	12.70	0.30	14.73	0.62
	LlamaGen	8 $\times$	4.39	17.86	65.97	29.41	19.69	44.56	82.76	49.00	10.51	0.18	20.85	0.68
O-MAGVIT2	8 $\times$	9.33	34.16	77.24	40.24	30.82	59.89	89.19	59.97	8.99	0.16	21.71	0.71	
Continuous	DC-AE	32 $\times$	1.42	5.16	42.45	16.35	10.95	24.06	66.45	33.82	14.61	0.22	20.42	0.67
	VA-VAE	16 $\times$	6.92	28.25	75.96	37.04	25.14	55.30	88.52	56.32	8.22	0.16	21.94	0.71
	SD-XL	8 $\times$	6.94	24.83	70.85	34.21	25.03	50.96	85.03	53.68	8.93	0.18	21.69	0.70
	SD-3.5	8 $\times$	36.26	72.18	92.68	67.04	59.04	85.64	97.06	80.58	8.40	0.13	23.46	0.75
	FLUX.1-dev	8 $\times$	50.69	82.14	94.89	75.91	70.70	90.67	97.90	86.42	7.19	0.12	23.93	0.76
Resolution: 512 $\times$ 512														
Discrete	Resize	1 $\times$	92.51	98.18	98.86	96.52	96.25	99.24	99.64	98.38	0.26	0.01	29.80	0.91
	VQGAN	16 $\times$	0.15	0.76	17.45	6.12	5.20	8.99	37.77	17.32	6.87	0.19	19.24	0.65
	Chameleon	16 $\times$	0.60	2.67	31.39	11.55	7.63	17.82	54.95	26.80	5.61	0.17	19.81	0.66
	LlamaGen	16 $\times$	0.67	3.93	40.43	15.01	7.76	20.17	63.39	30.44	5.28	0.15	20.21	0.68
	VAR	16 $\times$	3.71	20.59	63.62	29.31	18.01	49.56	82.44	50.00	3.78	0.12	21.27	0.73
	TokenFlow	16 $\times$	1.06	6.27	44.88	17.40	10.00	28.39	68.07	35.49	5.14	0.15	20.46	0.68
	O-MAGVIT2	16 $\times$	1.40	9.51	54.04	21.65	10.79	33.15	74.94	39.63	3.65	0.13	21.11	0.71
	O-MAGVIT2(pretrain)	16 $\times$	3.02	16.25	62.72	27.33	16.87	44.50	80.48	47.28	3.51	0.12	21.54	0.72
	UniTok	16 $\times$	17.25	51.86	81.20	50.10	44.75	76.28	92.20	71.08	3.27	0.09	22.54	0.76
	OmniTokenizer	8 $\times$	6.21	38.33	82.91	42.48	23.44	65.99	92.66	60.70	5.67	0.20	15.00	0.67
	LlamaGen	8 $\times$	12.13	56.66	88.89	52.56	34.11	77.45	95.39	68.99	2.60	0.07	23.46	0.78
O-MAGVIT2	8 $\times$	20.66	70.36	90.66	60.56	46.60	85.41	96.00	76.00	2.34	0.07	24.39	0.80	
Continuous	DC-AE	32 $\times$	5.31	30.10	79.33	38.25	20.91	57.78	89.98	56.22	2.33	0.09	23.24	0.76
	VA-VAE	16 $\times$	12.72	58.73	88.43	53.30	34.80	78.86	95.30	69.65	2.23	0.07	24.07	0.79
	SD-XL	8 $\times$	16.53	62.87	91.20	56.86	40.43	80.83	96.40	72.55	2.00	0.06	24.67	0.80
	SD-3.5	8 $\times$	56.55	91.64	97.33	81.84	75.56	96.44	98.91	90.30	1.33	0.03	26.57	0.85
	FLUX.1-dev	8 $\times$	70.29	94.62	98.02	87.64	84.67	97.65	99.26	93.86	0.73	0.03	27.25	0.86
Resolution: 1024 $\times$ 1024														
Discrete	Resize	1 $\times$	95.15	98.39	99.30	97.61	97.97	99.33	99.77	99.02	0.18	0.01	inf	0.96
	VQGAN	16 $\times$	0.76	2.69	41.53	15.00	7.90	15.03	63.47	28.80	4.03	0.11	21.67	0.74
	Chameleon	16 $\times$	3.00	8.22	59.33	23.52	14.46	29.14	77.56	40.39	2.98	0.09	22.33	0.75
	LlamaGen	16 $\times$	3.30	10.62	67.63	27.19	14.13	33.02	83.57	43.58	3.35	0.09	22.74	0.77
	VAR	16 $\times$	9.64	30.08	75.35	38.36	29.07	59.16	89.51	59.25	4.85	0.10	22.40	0.79
	TokenFlow	16 $\times$	4.46	14.86	68.57	29.30	18.62	41.21	84.43	48.09	3.34	0.09	23.26	0.78
	O-MAGVIT2	16 $\times$	5.76	20.74	77.71	34.74	19.42	47.04	89.63	52.03	2.47	0.07	23.86	0.80
	O-MAGVIT2(pretrain)	16 $\times$	9.08	29.35	79.77	39.40	28.65	57.43	90.78	58.95	2.32	0.07	24.46	0.81
	UniTok	16 $\times$	26.90	47.91	74.29	49.70	54.36	72.48	87.93	71.59	4.02	0.07	24.22	0.83
	OmniTokenizer	8 $\times$	14.27	54.67	91.49	53.48	36.63	76.92	96.53	70.02	4.13	0.16	15.30	0.74
	LlamaGen	8 $\times$	25.42	71.63	94.61	63.89	50.33	86.45	97.95	78.24	1.74	0.04	26.57	0.86
	O-MAGVIT2	8 $\times$	35.29	78.91	94.84	69.68	60.97	90.36	98.03	83.12	2.21	0.06	27.07	0.88
	Continuous	DC-AE	32 $\times$	15.32	48.36	92.72	52.14	35.47	71.69	96.89	68.02	1.11	0.04	27.01
VA-VAE		16 $\times$	25.14	69.54	93.84	62.84	48.94	85.17	97.52	77.21	1.59	0.04	27.31	0.87
SD-XL		8 $\times$	31.41	75.83	96.29	67.84	56.60	88.34	98.52	81.15	1.01	0.03	28.60	0.88
SD-3.5		8 $\times$	74.88	95.76	98.50	89.71	87.57	98.15	99.44	95.05	0.54	0.02	29.80	0.92
FLUX.1-dev		8 $\times$	83.71	96.83	98.72	93.09	92.68	98.69	99.52	96.96	0.41	0.01	30.55	0.94

Table 4: Performance of discrete and continuous tokenizer on TokBench face-set.

Type	Method	Factor	Similarity↑				rFID↓	LPIPS↓	PSNR↑	SSIM↑
			Small	Medium	Large	Mean				
Resolution: 256 × 256										
Discrete	Resize	1×	0.85	0.97	0.98	0.93	7.83	0.05	29.83	0.87
	TiTok	1D	0.03	0.03	0.05	0.04	23.11	0.53	13.31	0.43
	FlexTok	1D	0.06	0.12	0.25	0.15	13.54	0.38	17.18	0.54
	VQGAN	16×	0.05	0.08	0.17	0.10	18.08	0.38	17.39	0.52
	Chameleon	16×	0.08	0.15	0.30	0.18	25.87	0.39	17.94	0.53
	LlamaGen	16×	0.07	0.11	0.26	0.15	15.30	0.32	18.38	0.55
	VAR	16×	0.10	0.20	0.41	0.23	13.11	0.25	20.20	0.61
	MaskBit	16×	0.06	0.09	0.19	0.11	15.92	0.39	18.23	0.55
	TokenFlow	16×	0.07	0.13	0.26	0.15	13.43	0.30	18.85	0.56
	O-MAGVIT2	16×	0.08	0.15	0.34	0.19	12.91	0.29	19.24	0.58
	O-MAGVIT2(pretrain)	16×	0.08	0.16	0.35	0.20	12.92	0.29	19.49	0.59
	UniTok	16×	0.15	0.32	0.58	0.35	11.25	0.21	21.66	0.65
	OmniTokenizer	8×	0.15	0.34	0.61	0.37	12.06	0.31	15.53	0.56
	LlamaGen	8×	0.17	0.38	0.66	0.40	12.01	0.20	22.09	0.66
	O-MAGVIT2	8×	0.23	0.48	0.74	0.48	11.47	0.18	23.27	0.69
Continuous	DC-AE	32×	0.10	0.21	0.45	0.26	17.58	0.25	21.30	0.62
	VA-VAE	16×	0.22	0.48	0.76	0.49	9.26	0.16	23.85	0.70
	SD-XL	8×	0.18	0.40	0.69	0.42	11.19	0.20	23.29	0.68
	SD-3.5	8×	0.43	0.76	0.92	0.70	9.91	0.13	26.20	0.75
	FLUX.1-dev	8×	0.52	0.83	0.95	0.76	9.32	0.11	26.94	0.78
Resolution: 512 × 512										
Discrete	Resize	1×	0.95	0.99	1.00	0.98	0.08	0.00	37.34	0.97
	VQGAN	16×	0.08	0.11	0.37	0.19	7.33	0.23	20.42	0.61
	Chameleon	16×	0.13	0.21	0.50	0.28	6.62	0.22	20.98	0.61
	LlamaGen	16×	0.11	0.17	0.48	0.25	5.28	0.18	21.41	0.65
	VAR	16×	0.14	0.24	0.57	0.32	4.59	0.15	22.16	0.69
	TokenFlow	16×	0.11	0.16	0.45	0.24	6.48	0.19	21.39	0.64
	O-MAGVIT2	16×	0.13	0.22	0.58	0.31	4.67	0.16	22.40	0.67
	O-MAGVIT2(pretrain)	16×	0.13	0.22	0.57	0.31	4.55	0.16	22.66	0.68
	UniTok	16×	0.22	0.36	0.74	0.44	3.95	0.11	24.34	0.74
	OmniTokenizer	8×	0.24	0.45	0.80	0.50	5.11	0.20	15.93	0.63
	LlamaGen	8×	0.28	0.49	0.83	0.53	2.73	0.08	25.49	0.77
	O-MAGVIT2	8×	0.35	0.58	0.88	0.61	2.80	0.07	26.81	0.80
Continuous	DC-AE	32×	0.16	0.29	0.71	0.39	2.81	0.11	25.08	0.73
	VA-VAE	16×	0.31	0.54	0.87	0.57	2.41	0.07	26.84	0.79
	SD-XL	8×	0.29	0.51	0.87	0.55	2.46	0.07	27.14	0.79
	SD-3.5	8×	0.61	0.84	0.98	0.81	1.20	0.03	30.06	0.87
	FLUX.1-dev	8×	0.71	0.89	0.98	0.86	0.71	0.02	31.06	0.90
Resolution: 1024 × 1024										
Discrete	Resize	1×	1.0	1.0	1.0	1.0	0.01	0.00	inf	1.00
	VQGAN	16×	0.10	0.19	0.47	0.25	4.27	0.13	23.98	0.72
	Chameleon	16×	0.19	0.30	0.58	0.36	3.63	0.11	24.54	0.73
	LlamaGen	16×	0.15	0.27	0.57	0.33	3.45	0.10	24.77	0.76
	VAR	16×	0.23	0.34	0.62	0.40	6.21	0.12	23.67	0.77
	TokenFlow	16×	0.14	0.26	0.55	0.31	4.63	0.10	25.00	0.76
	O-MAGVIT2	16×	0.20	0.34	0.66	0.40	3.62	0.09	26.01	0.79
	O-MAGVIT2(pretrain)	16×	0.21	0.34	0.65	0.40	3.48	0.09	26.12	0.79
	UniTok	16×	0.33	0.50	0.76	0.53	3.78	0.07	26.54	0.84
	OmniTokenizer	8×	0.40	0.60	0.82	0.61	4.63	0.15	16.00	0.71
	LlamaGen	8×	0.49	0.66	0.87	0.67	2.11	0.04	28.89	0.87
	O-MAGVIT2	8×	0.57	0.74	0.91	0.74	2.11	0.05	29.92	0.89
Continuous	DC-AE	32×	0.27	0.45	0.78	0.50	1.44	0.05	29.48	0.84
	VA-VAE	16×	0.49	0.68	0.89	0.69	2.38	0.04	30.69	0.88
	SD-XL	8×	0.50	0.69	0.91	0.70	1.25	0.03	31.39	0.89
	SD-3.5	8×	0.86	0.94	0.99	0.93	0.42	0.01	33.07	0.96
	FLUX.1-dev	8×	0.92	0.97	0.99	0.96	0.24	0.01	33.61	0.97

Table 5: Performance of video tokenizer on TokBench-Video. The resolution refers specifically to the shorter edge of the videos, while maintaining the original aspect ratio throughout. The categorization into small, medium, and large scales is dynamically adjusted based on resolution.

Type	Method	Factor	T-ACC(%)↑				T-NED(%)↑				F-Sim↑			
			Small	Medium	Large	Mean	Small	Medium	Large	Mean	Small	Medium	Large	Mean
Resolution: 256														
Discrete	Resize	1 × 1 × 1	76.09	92.14	96.18	88.14	85.77	95.79	98.32	93.29	0.81	0.91	0.97	0.90
	Cosmos-VAE [15]	4 × 8 × 8	1.49	22.82	66.12	30.14	7.76	44.61	77.15	43.18	0.29	0.52	0.76	0.52
	Cosmos-VAE [15]	8 × 16 × 16	0.02	0.38	2.79	1.06	0.84	3.14	12.95	5.64	0.10	0.13	0.25	0.16
Continuous	Cosmos-VAE [15]	4 × 8 × 8	5.80	52.09	78.34	45.41	15.80	68.06	85.63	56.50	0.47	0.72	0.89	0.69
	Hunyuan-Video [7]	4 × 8 × 8	26.85	69.12	87.47	61.15	45.55	80.54	93.12	73.07	0.60	0.80	0.92	0.77
	CogVideoX [24]	4 × 8 × 8	24.80	72.47	86.34	61.21	43.06	82.29	92.41	72.59	0.58	0.78	0.91	0.76
	Cosmos-VAE [15]	8 × 16 × 16	0.45	6.23	48.99	18.56	3.25	24.62	64.08	30.65	0.21	0.39	0.65	0.42
	Step-Video [13]	8 × 16 × 16	17.39	61.40	82.41	53.73	33.16	75.67	89.76	66.19	0.48	0.69	0.86	0.67
Resolution: 480														
Discrete	Resize	1 × 1 × 1	64.44	90.74	96.92	84.04	77.71	95.72	98.57	90.67	0.82	0.89	0.95	0.89
	Cosmos-VAE [15]	4 × 8 × 8	0.90	20.32	73.71	31.64	6.74	41.81	83.53	44.03	0.44	0.60	0.80	0.61
	Cosmos-VAE [15]	8 × 16 × 16	0.02	0.90	13.82	4.91	0.85	4.20	27.02	10.69	0.19	0.18	0.31	0.23
Continuous	Cosmos-VAE [15]	4 × 8 × 8	5.30	46.80	86.82	46.31	14.99	64.63	92.20	57.27	0.60	0.77	0.90	0.76
	Hunyuan-Video [7]	4 × 8 × 8	28.65	64.49	91.83	61.66	44.43	77.83	95.83	72.70	0.69	0.82	0.92	0.81
	CogVideoX [24]	4 × 8 × 8	28.02	65.41	91.71	61.71	43.47	78.24	95.60	72.43	0.67	0.80	0.91	0.79
	Cosmos-VAE [15]	8 × 16 × 16	0.36	9.40	61.81	23.86	3.20	22.70	73.76	33.22	0.34	0.47	0.71	0.51
	Step-Video [13]	8 × 16 × 16	20.27	54.18	87.14	53.86	35.43	71.39	93.04	66.62	0.60	0.73	0.86	0.73

Table 6: Comparison of reconstruction quality (on TokBench and ImageNet) and ImageNet class-conditional generation performance. gFID results are cited from original publications using their respective generators. ‘-’ indicates unavailable results.

Method	TokBench Face		TokBench Text		ImageNet Rec.		ImageNet Gen.		Generator
	F-Sim _m ↑	Rank	T-ACC _m (%)↑	Rank	rFID↓	Rank	gFID↓	Rank	
VQGAN	0.07	9	1.10	9	7.94	10	15.78	8	MaskGIT
LlamaGen	0.10	8	4.28	8	2.19	8	3.06	7	LlamaGen-L
VAR	0.16	5	15.74	5	0.90	5	1.73	3	VAR-d30
TiTok	0.04	10	0.09	10	2.21	9	2.77	6	MaskGIT
TokenFlow	0.11	7	6.73	7	1.37	7	-	-	-
UniTok	0.23	4	44.59	1	0.41	3	2.51	5	LlamaGen-L
O-MAGVIT2	0.13	6	7.52	6	1.17	6	2.33	4	LlamaGen-L
O-MAGVIT2(F8)	0.32	1	40.24	2	0.34	2	-	-	-
VA-VAE	0.32	1	37.04	3	0.28	1	1.35	1	LlamaGen-XL
DC-AE	0.26	3	16.35	4	0.69	4	1.72	2	SiT-XL

make autoregressive generation computationally expensive, illustrating the trade-off between reconstruction fidelity and generation efficiency.

Our metrics provide complementary insights. While rFID captures global reconstruction quality, our T-ACC and F-Sim reveal fine-grained differences that rFID misses. For instance, UniTok and O-MAGVIT2 have comparable rFID (0.41 vs. 1.17) but differ dramatically in T-ACC (44.59% vs. 7.52%), indicating that UniTok’s multi-codebook quantization is substantially better at preserving text details. These distinctions are invisible to conventional metrics alone.

These observations confirm that while reconstruction quality provides a necessary foundation for generation fidelity, generation performance is additionally influenced by the generator architecture, training configuration, and the modeling difficulty of the latent

space. TokBench serves as an efficient tool for evaluating the reconstruction aspect of this equation.

B.5 Evaluation on Texture Reconstruction

To examine whether our benchmark’s findings generalize beyond text and faces, we conduct additional experiments on the Describable Textures Dataset (DTD), evaluating reconstruction performance using conventional metrics. As shown in Table 7, the overall ranking of tokenizers on textures is broadly consistent with our findings on faces and text, confirming the generalizability of our main conclusions. However, conventional metrics (rFID, LPIPS, PSNR, SSIM) yield inconsistent fine-grained rankings among top-performing models, highlighting the difficulty of reliable evaluation using generic metrics alone. This reinforces our motivation for focusing on text and faces, where task-aware metrics (OCR accuracy, face recognition similarity) provide objective and unambiguous

evaluation. Moreover, text inherently shares critical properties with fine textures and thin lines: dense text blocks can be perceived as a form of fine texture, and characters are fundamentally composed of thin strokes whose accurate reconstruction is directly captured by our OCR-based metrics.

Table 7: Texture reconstruction evaluation on DTD. Rankings are shown after each value. The overall trend is consistent with TokBench results, but conventional metrics show inconsistent fine-grained rankings among top models.

Method	TokBench		DTD Texture				
	T-ACC _m (%)↑	F-Sim _m ↑	rFID↓	LPIPS↓	PSNR↑	SSIM↑	
VQGAN	1.10 /14	0.07 /14	15.51 /14	0.37 /14	18.55 /14	0.35 /14	
Chameleon	2.87 /13	0.12 /11	13.67 /13	0.36 /13	18.98 /13	0.37 /13	
LlamaGen	4.28 /12	0.10 /13	11.75 /12	0.31 /12	19.47 /12	0.41 /11	
VAR	15.74 /8	0.16 /8	9.03 /8	0.25 /8	21.25 /8	0.50 /8	
TiTok	0.09 /15	0.04 /15	19.85 /15	0.51 /15	15.43 /15	0.25 /15	
TokenFlow	6.73 /11	0.11 /12	11.03 /11	0.29 /11	19.87 /11	0.41 /11	
UniTok	44.59 /3	0.23 /7	8.41 /6	0.21 /6	22.62 /7	0.56 /7	
O-MAGVIT2	7.52 /10	0.13 /9	9.47 /9	0.29 /9	20.34 /10	0.44 /10	
O-MAGVIT2(pt)	10.58 /9	0.13 /9	9.59 /10	0.29 /9	20.46 /9	0.45 /9	
LlamaGen(F8)	29.41 /7	0.26 /6	7.61 /5	0.19 /4	22.70 /6	0.57 /5	
O-MAGVIT2(F8)	40.24 /4	0.32 /3	7.43 /4	0.19 /4	23.63 /5	0.61 /3	
VA-VAE	37.04 /5	0.32 /3	6.16 /3	0.18 /3	23.99 /4	0.60 /4	
SD-XL	34.21 /6	0.27 /5	8.58 /7	0.21 /6	23.79 /3	0.57 /5	
SD-3.5	67.04 /2	0.49 /2	5.82 /2	0.14 /2	26.24 /2	0.70 /2	
FLUX.1-dev	75.91 /1	0.56 /1	5.26 /1	0.11 /1	27.20 /1	0.75 /1	

B.6 T-ACC vs. Text Feature Similarity

An alternative to using OCR recognition accuracy for text evaluation is to compute feature similarity from the OCR encoder, analogous to F-Sim for faces. To compare these two approaches, we extract features from the PARSeq encoder and compute cosine similarity between original and reconstructed text regions. As shown in Table 8, both metrics yield nearly identical tokenizer rankings, with only minor differences at rank 10–11 (O-MAGVIT2 and TokenFlow). However, T-ACC provides substantially better discriminability: the similarity scores are compressed into a narrow range (0.50–0.92), making it difficult to distinguish between tokenizers with similar performance. In contrast, T-ACC spans a much wider range (0.09%–75.91%), offering clearer separation. Furthermore, recognition accuracy aligns more naturally with human judgment, which is typically based on whether text is readable rather than on abstract feature distances.

C Ablation Study

C.1 Impact of Text-Rich Training Data

Since different tokenizers are typically trained on distinct datasets, a natural question is whether incorporating more text-rich data can improve text reconstruction quality. To investigate this, we conduct ablation studies following LlamaGen’s [18] training protocol. We train both F16 and F8 VQGAN models on two dataset configurations: (1) ImageNet [3] alone, and (2) ImageNet augmented with an additional 230K text-rich images. Both configurations are evaluated on the TokBench text set, with results shown in Table 9.

The results demonstrate that incorporating text-rich training data improves T-ACC and T-NED scores across both downsampling

Table 8: Comparison of T-ACC and text feature similarity. Both metrics yield consistent rankings, but T-ACC offers better discriminability across the full performance range.

Method	T-ACC _m (%)↑	Rank	Similarity↑	Rank
VQGAN	1.10	14	0.60	14
Chameleon	2.87	13	0.63	13
LlamaGen	4.28	12	0.65	12
VAR	15.74	8	0.74	8
TiTok	0.09	15	0.50	15
TokenFlow	6.73	11	0.69	10
UniTok	44.59	3	0.84	3
O-MAGVIT2	7.52	10	0.68	11
O-MAGVIT2(pt)	10.58	9	0.72	9
LlamaGen(F8)	29.41	7	0.79	7
O-MAGVIT2(F8)	40.24	4	0.83	4
VA-VAE	37.04	5	0.82	6
SD-XL	34.21	6	0.82	5
SD-3.5	67.04	2	0.90	2
FLUX.1-dev	75.91	1	0.92	1

Table 9: Ablations on Training Data. While augmenting ImageNet with text-rich data yields performance improvements, the gains remain modest compared to architectural differences observed in Table 1 of the main paper.

Method	Data	T-ACC _s ↑	T-ACC _m ↑	T-NED _s ↑	T-NED _m ↑
F16	ImageNet	0.02	2.96	4.84	12.11
F16	ImageNet+Text	0.09	3.93	5.19	14.48
F8	ImageNet	2.99	25.99	16.25	45.09
F8	ImageNet+Text	3.42	27.51	18.05	47.36

factors. However, these improvements are relatively marginal compared to the gains achieved through architectural enhancements such as advanced quantization methods (e.g., MCQ in UniTok) or reduced downsampling ratios (F8 vs. F16). This suggests that while training data composition influences reconstruction quality, tokenizer architecture and quantization design remain the more critical factors for preserving fine-grained visual details.

C.2 Training Details

Our ablation uses the ImageNet [3] training set as the baseline. For the augmented configuration, we supplement with 230,000 text-rich images sourced from the training sets of Synth150K [9], ICDAR 2017 MLT [14], Total-Text [2], TextOCR [17], CTW1500 [10], and COCO-Text [20]. Importantly, the additional text images are drawn from different splits or entirely different datasets than those used in TokBench evaluation, ensuring no data leakage. Only the images are used for training; no text annotations are required. To ensure a fair comparison, both configurations are trained for 400,000 iterations under identical hyperparameters, following the default LlamaGen VQGAN training recipe.

D More Visualization

Tables 1 and 2 present qualitative comparisons of reconstruction results from different methods at 256 and 1024 resolutions respectively. At 256 resolution, most discrete tokenizers fail to accurately

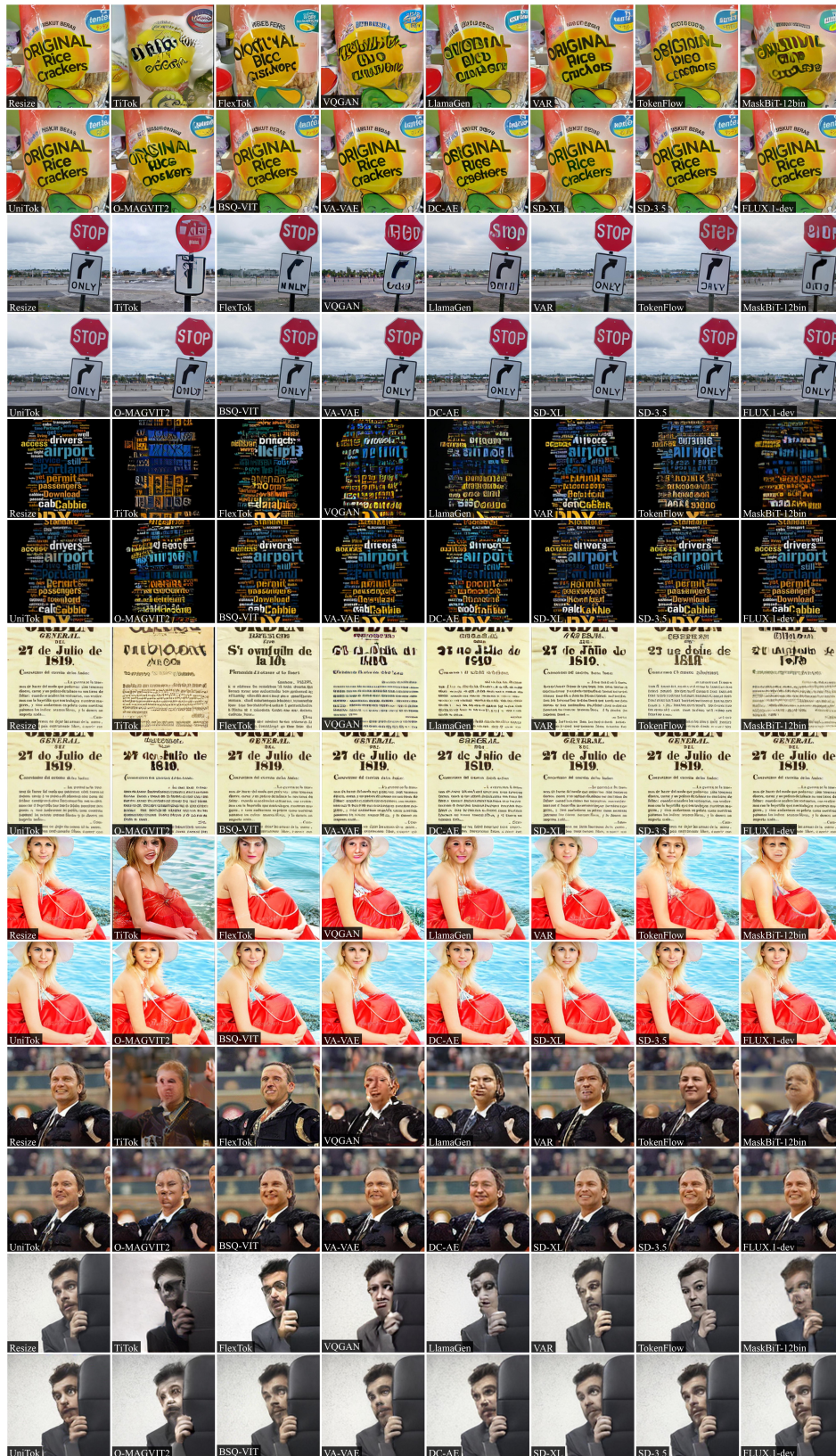


Figure 1: Visualization results of text and face reconstruction performance for different methods at 256 resolution. (Zoom in for better comparison.)

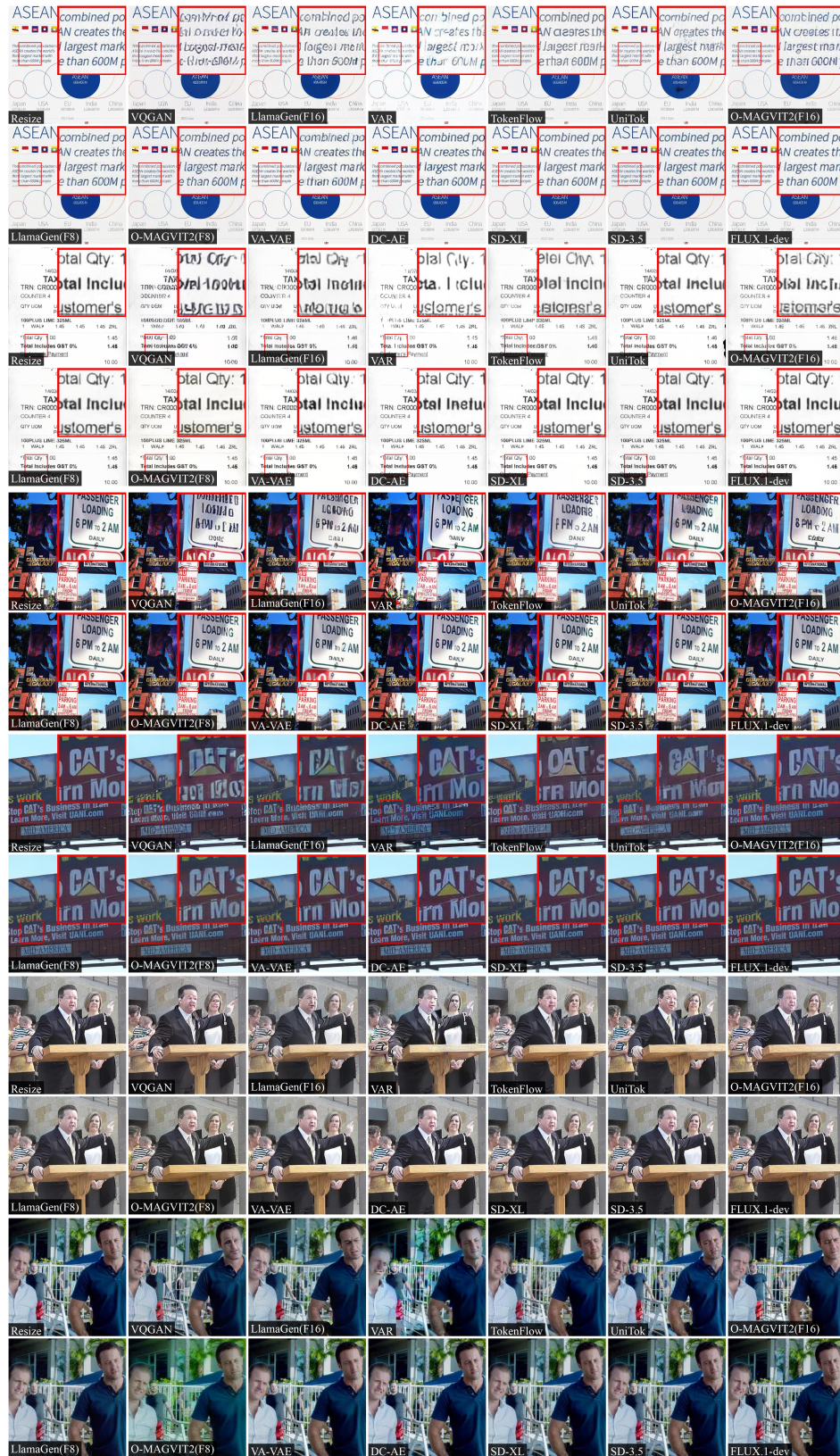


Figure 2: Visualization results of text and face reconstruction performance for different methods at 1024 resolution. (Zoom in for better comparison.)

reconstruct text and faces, while the high-compression DC-AE also performs poorly. In contrast, SD3.5 and FLUX VAEs demonstrate significantly better visual quality. At 1024 resolution, both VAEs and low-compression (F8) discrete tokenizers achieve satisfactory results, though F8 Open-MAGVIT2 exhibits noticeable color distortion, and F16 discrete tokenizers still struggle with small-scale objects.

References

- [1] Roman Bachmann, Jesse Allardice, David Mizrahi, Enrico Fini, Oğuzhan Fatih Kar, Elmira Amirloo, Alaaeldin El-Nouby, Amir Zamir, and Afshin Dehghan. 2025. FlexTok: Resampling Images into 1D Token Sequences of Flexible Length. *arXiv preprint arXiv:2025.00000* (2025).
- [2] Chee-Kheng Chng, Chee Seng Chan, and Cheng-Lin Liu. 2020. Total-Text: toward orientation robustness in scene text detection. *Int. J. Document Anal. Recognit.* 23, 1 (2020), 31–52. doi:10.1007/S10032-019-00334-Z
- [3] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*. 248–255.
- [4] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. 2024. Scaling rectified flow transformers for high-resolution image synthesis. In *International Conference on Machine Learning*.
- [5] Patrick Esser, Robin Rombach, and Bjorn Ommer. 2021. Taming transformers for high-resolution image synthesis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 12873–12883.
- [6] Alex Fang, Albin Madappally Jose, Amit Jain, Ludwig Schmidt, Alexander Tshv, and Vaishal Shankar. 2023. Data filtering networks. *arXiv preprint arXiv:2309.17425* (2023).
- [7] Weijie Kong, Qi Tian, Zijian Zhang, Rox Min, Zuozhuo Dai, Jin Zhou, Jiangfeng Xiong, Xin Li, Bo Wu, Jianwei Zhang, et al. 2024. Hunyuanvideo: A systematic framework for large video generative models. *arXiv preprint arXiv:2412.03603* (2024).
- [8] Black Forest Labs. 2024. FLUX. <https://github.com/black-forest-labs/flux>.
- [9] Yuliang Liu, Hao Chen, Chunhua Shen, Tong He, Lianwen Jin, and Liangwei Wang. 2020. Abcnet: Real-time scene text spotting with adaptive bezier-curve network. In *proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9809–9818.
- [10] Yuliang Liu, Lianwen Jin, Shuaitao Zhang, Canjie Luo, and Sheng Zhang. 2019. Curved scene text detection via transverse and longitudinal sequence connection. *Pattern Recognition* 90 (2019), 337–345. doi:10.1016/j.patcog.2019.02.002
- [11] Zhuoyan Luo, Fengyuan Shi, Yixiao Ge, Yujiu Yang, Limin Wang, and Ying Shan. 2024. Open-magvit2: An open-source project toward democratizing autoregressive visual generation. *arXiv preprint arXiv:2409.04410* (2024).
- [12] Chuofan Ma, Yi Jiang, Junfeng Wu, Jihan Yang, Xin Yu, Zehuan Yuan, Bingyue Peng, and Xiaojuan Qi. 2025. UniTok: A Unified Tokenizer for Visual Generation and Understanding. *arXiv preprint arXiv:2502.20321* (2025).
- [13] Guoqing Ma, Haoyang Huang, Kun Yan, Liangyu Chen, Nan Duan, Shengming Yin, Changyi Wan, Ranchen Ming, Xiaoni Song, Xing Chen, et al. 2025. Step-video-t2v technical report: The practice, challenges, and future of video foundation model. *arXiv preprint arXiv:2502.10248* (2025).
- [14] Nibal Nayef, Fei Yin, Imen Bizid, Hyunsoo Choi, Yuan Feng, Dimosthenis Karatzas, Zhenbo Luo, Umapada Pal, Christophe Rigaud, Joseph Chazalon, et al. 2017. Icdar2017 robust reading challenge on multi-lingual scene text detection and script identification-rrc-mlt. In *International Conference on Document Analysis and Recognition*, Vol. 1. 1454–1459.
- [15] NVIDIA. 2024. Cosmos-tokenizer. <https://research.nvidia.com/labs/dir/cosmos-tokenizer/>.
- [16] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. 2023. Sdxl: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952* (2023).
- [17] Amanpreet Singh, Guan Pang, Mandy Toh, Jing Huang, Wojciech Galuba, and Tal Hassner. 2021. TextOCR: Towards Large-Scale End-to-End Reasoning for Arbitrary-Shaped Scene Text. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8802–8812. doi:10.1109/CVPR46437.2021.00869
- [18] Peize Sun, Yi Jiang, Shoufa Chen, Shilong Zhang, Bingyue Peng, Ping Luo, and Zehuan Yuan. 2024. Autoregressive model beats diffusion: Llama for scalable image generation. *arXiv preprint arXiv:2406.06525* (2024).
- [19] Keyu Tian, Yi Jiang, Zehuan Yuan, Bingyue Peng, and Liwei Wang. 2024. Visual autoregressive modeling: Scalable image generation via next-scale prediction. *Advances in neural information processing systems* 37 (2024), 84839–84865.
- [20] Andreas Veit, Tomas Matera, Lukas Neumann, Jiri Matas, and Serge Belongie. 2016. Coco-text: Dataset and benchmark for text detection and recognition in natural images. *arXiv preprint arXiv:1601.07140* (2016).
- [21] Junke Wang, Yi Jiang, Zehuan Yuan, Bingyue Peng, Zuxuan Wu, and Yu-Gang Jiang. 2024. OmniTokenizer: A Joint Image-Video Tokenizer for Visual Generation. In *Advances in Neural Information Processing Systems*, Vol. 37. 28281–28295.
- [22] Mark Weber, Lijun Yu, Qihang Yu, Xueqing Deng, Xiaohui Shen, Daniel Cremers, and Liang-Chieh Chen. 2024. MaskBit: Embedding-free Image Generation via Bit Tokens. *Transactions on Machine Learning Research* (2024). <https://openreview.net/forum?id=NYe2JuN3v3>
- [23] Enze Xie, Junsong Chen, Junyu Chen, Han Cai, Haotian Tang, Yujun Lin, Zhekai Zhang, Muyang Li, Ligeng Zhu, Yao Lu, et al. 2024. Sana: Efficient high-resolution image synthesis with linear diffusion transformers. *arXiv preprint arXiv:2410.10629* (2024).
- [24] Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Shiyu Huang, Jiazheng Xu, Yuanming Yang, Wenqi Hong, Xiaohan Zhang, Guanyu Feng, et al. 2024. CogVideoX: Text-to-Video Diffusion Models with An Expert Transformer. *arXiv preprint arXiv:2408.06072* (2024).
- [25] Jingfeng Yao, Bin Yang, and Xinggang Wang. 2025. Reconstruction vs. generation: Taming optimization dilemma in latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [26] Qihang Yu, Mark Weber, Xueqing Deng, Xiaohui Shen, Daniel Cremers, and Liang-Chieh Chen. 2024. An image is worth 32 tokens for reconstruction and generation. *Advances in Neural Information Processing Systems* 37 (2024), 128940–128966.